



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Power

Energy per second. Two equations, and a way to check you have used the right one.

Key concept

Power is how fast energy is transferred.

A circuit transfers energy — you have seen that all term. **Power** is the *rate*: the energy transferred each second.

A **60 W** lamp transfers **60 joules every second**. A **2000 W** kettle transfers **2000 joules every second**, which is why it boils water in a couple of minutes and the lamp never will.

Power is measured in **watts (W)**. 1 watt = 1 joule per second.

The two equations — learn both

Neither of these is on the equation sheet. You are expected to recall them and use them.

$$P = VI$$

power = potential difference $\times$ current

$$P = I^2 R$$

power = (current)² $\times$ resistance

The units, every time: P in **watts (W)**, V in **volts (V)**, I in **amps (A)**, R in **ohms (Ω)**

Which one do I use? Look at what the question gives you.

You are given V and I

use $P = VI$

You are given I and R

use $P = I^2 R$

You are given two of V , I , R

get the third from $V = IR$, then either works

Worked example 1 — you have V and I

A lamp has a potential difference of 6.0 V across it and a current of 0.50 A through it. Calculate the power of the lamp.

1. Pick V and I are given, so use $P = VI$

2. Substitute $P = 6.0 \times 0.50$

3. Answer $P = 3.0 \text{ W}$

Worked example 2 — you have I and R

A heater element has a resistance of 20 Ω and a current of 3.0 A through it. Calculate its power.

1. Pick I and R are given, so use $P = I^2 R$

2. Square first $I^2 = 3.0^2 = 9.0$

3. Then multiply $P = 9.0 \times 20 = 180 \text{ W}$

The check. If a question gives you enough to use *both* equations, use both. They describe the same energy going through the same component, so they give the **same number**. Matching answers tell you the substitution was right — question 4 is built to show you this.

Questions

Show the equation, the substitution and the answer with its unit on every one.

- 1) **Calculate** the power of a lamp that has a potential difference of 12 V across it and a current of 2.0 A through it. [2]

$$P = V I = 12 \times 2.0 = \mathbf{24 \text{ W}}$$

- 2) **Calculate** the power of a 15Ω resistor carrying a current of 0.40 A. [2]

$$P = I^2 R = 0.40^2 \times 15 = 0.16 \times 15 = \mathbf{2.4 \text{ W}}$$

- 3) **Calculate** the current in a 2300 W kettle connected to the 230 V mains supply. [3]

$$\text{Rearrange } P = V I \text{ to give } I = \frac{P}{V} = \frac{2300}{230} = \mathbf{10 \text{ A}}$$

- 4) A resistor of resistance 6.0Ω carries a current of 1.5 A.

- (a) Calculate the power of the resistor using $P = I^2 R$.

$$P = I^2 R = 1.5^2 \times 6.0 = 2.25 \times 6.0 = \mathbf{13.5 \text{ W}}$$

[2]

- (b) Now find the potential difference across the resistor, and use it to work out the power a second way.

$$V = I R = 1.5 \times 6.0 = 9.0 \text{ V, then } P = V I = 9.0 \times 1.5 = \mathbf{13.5 \text{ W}} \text{ — the same answer, which confirms both substitutions.}$$

[3]